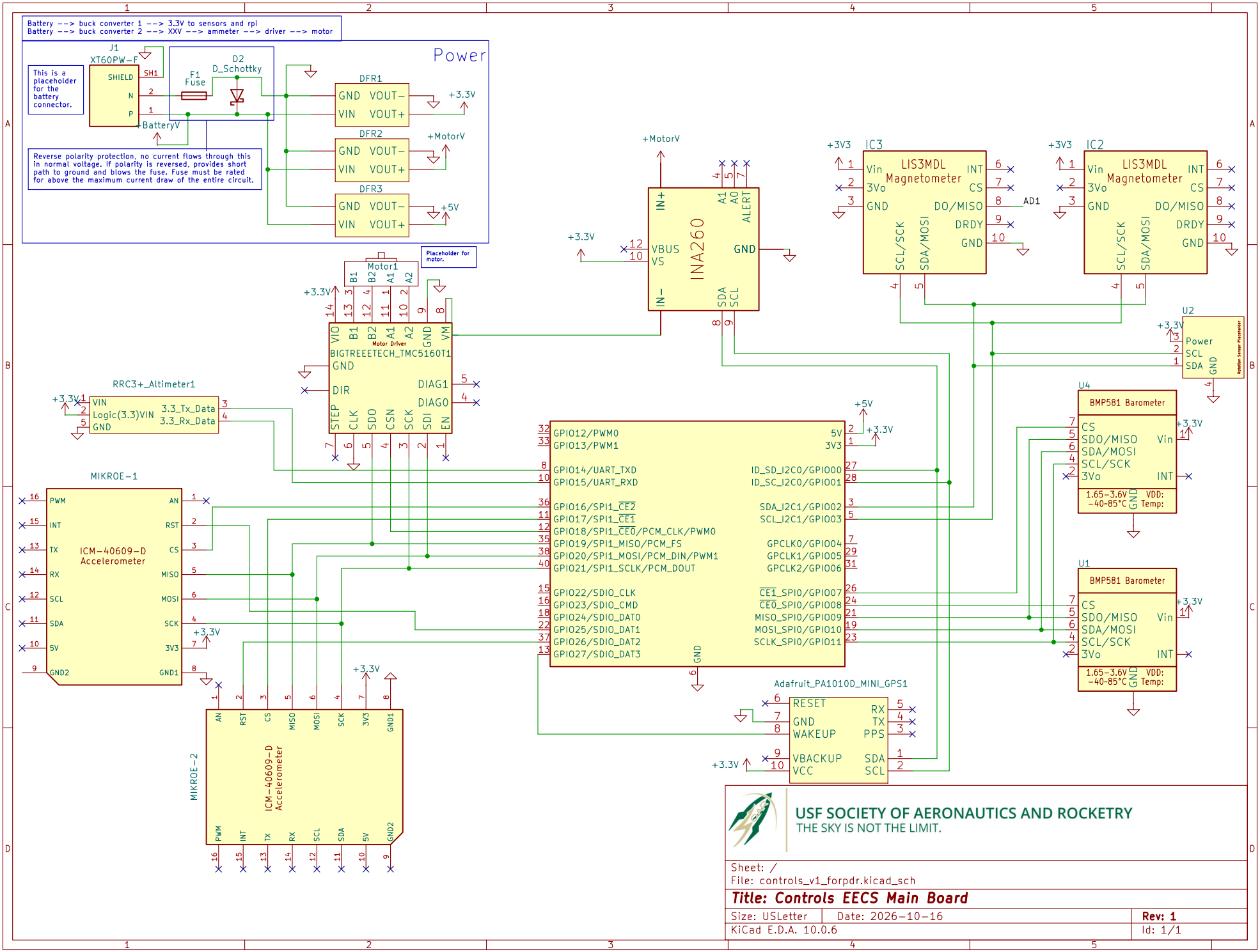

Battery --> buck converter 1 --> 3.3V to sensors and rpi
Battery --> buck converter 2 --> XXV --> ammeter --> driver --> motor
Power
This is a placeholder for the battery connector.
J1
XT60PW-F
SHIELD
SH1
N
2
P
1
+BatteryV
F1
Fuse
D2
D_Schottky
Reverse polarity protection, no current flows through this in normal voltage. If polarity is reversed, provides short path to ground and blows the fuse. Fuse must be rated for above the maximum current draw of the entire circuit.
DFR1
GND VOUT-
VIN VOUT+
+3.3V
DFR2
GND VOUT-
VIN VOUT+
+MotorV
DFR3
GND VOUT-
VIN VOUT+
+5V
Placeholder for motor.
Motor1
B1 B2 A1 A2
+3.3V
VIO B1 B2 A1 A2 GND VM
Motor Driver
BIGTREEETECH_TMC5160T1
GND
DIR
DIAG1
DIAG0
STEP CLK SDO CSN SCK SDI EN
+MotorV
INA260
IN+
IN-
VBUS
VS
A1 A0 ALERT
SDA SCL
GND
+3.3V
IC3
LIS3MDL
Magnetometer
Vin
3Vo
GND
SCL/SCK
SDA/MOSI
INT
CS
DO/MISO
AD1
DRDY
GND
IC2
LIS3MDL
Magnetometer
U2
Power
SCL
SDA
GND
Rotation Sensor Placeholder
RRC3+_Altimeter1
VIN
Logic(3.3)VIN
GND
3.3_Tx_Data
3.3_Rx_Data
MIKROE-1
ICM-40609-D
Accelerometer
PWM
INT
TX
RX
SCL
SDA
5V
GND2
AN
RST
CS
MISO
MOSI
SCK
3V3
GND1
MIKROE-2
ICM-40609-D
Accelerometer
GPIO12/PWM0
GPIO13/PWM1
GPIO14/UART_TXD
GPIO15/UART_RXD
GPIO16/SPI1_CE2
GPIO17/SPI1_CE1
GPIO18/SPI1_CE0/PCM_CLK/PWM0
GPIO19/SPI1_MISO/PCM_FS
GPIO20/SPI1_MOSI/PCM_DIN/PWM1
GPIO21/SPI1_SCLK/PCM_DOUT
GPIO22/SDIO_CLK
GPIO23/SDIO_CMD
GPIO24/SDIO_DAT0
GPIO25/SDIO_DAT1
GPIO26/SDIO_DAT2
GPIO27/SDIO_DAT3
5V
3V3
ID_SD_I2C0/GPIO00
ID_SC_I2C0/GPIO01
SDA_I2C1/GPIO02
SCL_I2C1/GPIO03
GPCLK0/GPIO04
GPCLK1/GPIO05
GPCLK2/GPIO06
CE1_SPI0/GPIO07
CE0_SPI0/GPIO08
MISO_SPI0/GPIO09
MOSI_SPI0/GPIO10
SCLK_SPI0/GPIO11
GND
Adafruit_PA1010D_MINI_GPS1
RESET
GND
WAKEUP
VBACKUP
VCC
RX
TX
PPS
SDA
SCL
U4
BMP581 Barometer
CS
SDO/MISO
SDA/MOSI
SCL/SCK
3Vo
Vin
INT
1.65-3.6V
-40-85°C
VDD:
Temp:
U1
BMP581 Barometer
USF SOCIETY OF AERONAUTICS AND ROCKETRY
THE SKY IS NOT THE LIMIT.
Sheet: /
File: controls_v1_forpdr.kicad_sch
Title: Controls EECS Main Board
Size: USLetter
Date: 2026-10-16
Rev: 1
KiCad E.D.A. 10.0.6
Id: 1/1